UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

| | |
|---|---|
| COURSE | PROGRAMMING I |
| COURSE CODE | CSC402 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 6 printed pages.*

«:> Hak Cipta Universiti Teknologi MARA

## PART A (10 MARKS)

*Answer ALL questions.*

1. Which data type stores a whole number?
   A. int
   B. char
   C. string
   D. float

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 1357
   B. 246
   C. 1234
   D. 2468

3. Which item is NOT a programming paradigm?
   A. Procedural programming
   B. Debugging
   C. Structured programming
   D. Object-oriented programming

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. n
   B. i
   C. t
   D. p

5. Which of the following are stream operators?
   i. <<
   ii. >>
   iii. ==
   iv. =
   A. i and ii
   B. iii and iv
   C. ii and iv
   D. i and iii

6. Which structure chooses between two alternative paths?
   A. for
   B. struct
   C. array
   D. if...else

7. Which tool represents an algorithm using standard graphical symbols?
   A. Compiler
   B. Debugger

**C** Hak Cipta Universiti Teknologi MARA

C. Pseudocode
D. Flowchart

8. What does the % operator return?
A. The remainder of integer division
B. The quotient
C. The absolute value
D. The square root

9. Which of the following declarations are valid?
i. int a[5];
ii. int b;
iii. int c[3] = {1, 2, 3};
iv. int d[];
A. i and ii
B. i and iii
C. ii and iv
D. iii and iv

10. Which statement is most suitable for selecting among several constant cases?
A. for
B. do...while
C. switch
D. while

## PART B (60 MARKS)

### QUESTION 1

This question tests algorithms and programming errors.

**a)** Define algorithm and pseudocode.

*(3 marks)*

**b)** Explain THREE types of programming errors and give one example of each.

*(2 marks)*

### QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

*Table 1. Concert Ticket Prices*

| Age Range | Ticket Price (RM) |
|---|---|
| 0 - 12 | 15.00 |
| 13 - 17 | 25.00 |
| 18 and above | 40.00 |

**a)** Draw a flowchart for the ticket price selection.

*(5 marks)*

**b)** Write pseudocode and a complete C++ program for the ticket price selection.

*(8 marks)*

### QUESTION 3

This question tests declarations, constants and expressions.

**a)** Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

*(6 marks)*

**b)** Write C++ statements to read age and price, calculate price after discount, and display the result.

*(5 marks)*

### QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

**a)** Trace the program and show the exact output.

*(10 marks)*

**b)** Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

*(5 marks)*

## QUESTION 5

A sports club records the sport names and medal counts in Table 1.

**a)** Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

*Table 2. Sport and Medal Records*

| Sport | Medals |
|---|---|
| Swimming | 8 |
| Badminton | 12 |
| Athletics | 7 |
| Cycling | 5 |
| Hockey | 9 |

*(6 marks)*

**b)** Write a program segment to display each sport with its medals, then calculate the total and average medals.

*(10 marks)*

## PART C (30 MARKS)

### QUESTION 1

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

*Table 3. Cafe Menu*

| Item | Price (RM) |
|---|---|
| Coffee | 8.50 |
| Tea | 6.00 |
| Juice | 9.50 |

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

*(15 marks)*

### QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

*Table 4. Book Record Fields*

| Title | Price (RM) | Quantity |
|---|---|---|
| Book A | 45.00 | 3 |
| Book B | 30.00 | 5 |
| Book C | 55.00 | 2 |

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

*(15 marks)*

*END OF QUESTION PAPER*